# Carryover Monitor Executive Summary

## Team THD | Rockwell Automation and NVIDIA Physical AI Hackathon | CSI Fluid Mixing Station 3

Team THD developed an advisory carryover monitor for shared-path fluid filling. The system combines recipe context with real-time vial imagery to warn operators before a risky recipe transition and to check whether the filled vial shows visible carryover. Our prototype demonstrates that this approach can run in real time on an NVIDIA DGX Spark, while also identifying the additional real-line validation and integration needed before production use.

Manufacturing problem and goal

Several source tanks feed a shared dispensing path. When the recipe changes, residual liquid can enter the first vial of the next batch and alter its color or composition. A missed defect may cause scrap, rework, added inspection, downstream quality problems, or an unnecessary line interruption. Our goal was to give operators early, explainable evidence while keeping the monitor read-only and advisory.

Physical AI approach

The process model compares the current and upcoming recipes to identify transitions with carryover risk. A compact neural network then evaluates the vial's developing color from camera frames, while interpretable color rules provide confirmation evidence. The dashboard runs on the DGX Spark and shows the active recipe, upcoming queue, live camera frame, model probability, and event history. A rolling three-of-five-frame rule reduces reactions to isolated noisy frames. The monitor recommends inspection, diversion, or an approved flush; it does not change PLC recipes or operate the line.

Data and use of CSI equipment

We used CSI digital-twin assets and a reproducible pipeline to create 480 labeled, 30 fps vial-fill scenarios at approximately live speed. Scenarios varied recipes, liquid colors, expected hue, carryover severity, and imaging conditions, with synchronized video, queue, and frame metadata. We also evaluated Dataset 5 footage from the real CSI vial-fill camera. Its supplied timing labels drifted from the visible fills, so we detected the fill periods from the images and corrected the alignment. The Spark executed the frozen model and dashboard replay at original timing. Direct PLC queue data was unavailable, so the hackathon queue was prepared manually; production would use an approved read-only PLC or MES feed. We did not complete an untouched live trial: the PLC-controlled machine was unavailable during our first visit, and authorized access to the line-connected Spark and camera was unavailable during the second.

Measured evidence

| Scored vials | Carryover detected | False alarms | ROC AUC | Mean inference |
|---|---|---|---|---|
| 23 | 6 of 7 | 2 | 0.839 | 4.1 ms |

At the balanced evaluation threshold of 0.0265, the model detected six of seven expected carryover cases, missed one, and produced two false alarms, for 85.7% recall, 87.5% specificity, 75% precision, and approximately 87% accuracy. The demonstration dashboard uses a separate sensitivity setting, so the balanced-threshold accuracy figures should not be assigned to its warnings.

Manufacturing value and deployment path

Earlier warning and visual confirmation could reduce defect escapes, manual inspection, rework, and troubleshooting time. Recipe-aware monitoring may also support targeted flushing instead of flushing after every change, reducing material waste and downtime. The concept applies to paint, chemical, pharmaceutical, food and beverage, and personal-care processes that share dispensing paths. Production deployment would require authorized live-camera access, more labeled real fills, threshold calibration on separate validation data, and a final untouched test set.

Supporting material https://github.com/ALL-Applied-AI-Network/msoe-ai-club-rockwell-automation-nvidia-hackath-thd-16c769df76/tree/main